

Exercise 9.1 - Observability

Course: Optimizaciones y Desempeño — Cloud Deployment Automation

Session: 9 — June 18, 2026

Time allowed: 30 minutes

Submission: Initialize a new repository called oyd-exercise-9-1 and commit/push everything into it. Submit the repository URL only.

Context

FinAPI is a fintech startup that processes payment requests behind an Application Load Balancer on AWS. The team has networking and compute already deployed via Terraform modules. With a production release scheduled for next week, the engineering lead has asked you to add a complete observability module: a CloudWatch log group, error-rate and latency alarms connected to an SNS email subscription, an EstimatedCharges alarm that fires near-real-time, and a monthly budget guard.

Your starter project has the following root files. Clone or copy them as your starting point.

versions.tf

```
terraform {
  required_version = ">= 1.6"
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = var.aws_region
}
```

variables.tf

```
variable "aws_region" {
  description = "AWS region for all resources"
  type        = string
  default     = "us-east-2"
}

variable "alb_arn_suffix" {
  description = "ARN suffix of your ALB (from the AWS console, e.g. app/my-alb/abc123def456)"
}
```

```

    type          = string
}

variable "notification_email" {
    description = "Email address that receives CloudWatch alarm notifications"
    type        = string
}

variable "log_retention_days" {
    description = "Days to retain log group entries"
    type        = number
    default     = 14
}

variable "monthly_budget_usd" {
    description = "Monthly budget ceiling in USD"
    type        = number
    default     = 25
}

variable "estimated_charges_threshold" {
    description = "USD amount that triggers the EstimatedCharges alarm. Set to
a value your account has already exceeded before class so the alarm fires and
you can see it in the console."
    type        = number
    default     = 10
}

```

main.tf

```
# Task 5: call the observability module here
```

outputs.tf

```
# Outputs to be added after the observability module is wired in
```

envs/dev/dev.tfvars

```

aws_region          = "us-east-2"
alb_arn_suffix      = "app/<YOUR-ALB-NAME>/<YOUR-ALB-ID>"
notification_email   = "<YOUR-EMAIL>"
log_retention_days   = 14
monthly_budget_usd   = 25
estimated_charges_threshold = 10

```

Setup

Prerequisites

- AWS CLI configured with valid credentials (aws sts get-caller-identity should return your account ID)
- Terraform >= 1.6 installed
- Your team's ALB ARN suffix — find it in the AWS console under EC2 > Load Balancers; copy the value shown as "ARN suffix" on the detail page
- An email address you can check during the exercise — AWS will send an SNS subscription confirmation you must click

Repository structure

```
oyd-exercise-9-1/
├── versions.tf
├── main.tf
├── variables.tf
├── outputs.tf
├── envs/
│   └── dev/
│       └── dev.tfvars
├── modules/
│   └── observability/
│       ├── main.tf      ← create this
│       ├── variables.tf ← create this
│       └── outputs.tf   ← create this
```

Tasks

Task 1 — Log group and SNS

Create modules/observability/main.tf. Add the following resources:

- aws_cloudwatch_log_group — name /finapi/dev, retention controlled by a variable (default 14 days). Use name_prefix if you want to avoid name conflicts.
- aws_sns_topic — a single topic for alarm notifications (e.g. name = "finapi-alerts").
- aws_sns_topic_subscription — protocol = "email", endpoint = the notification email variable, topic_arn referencing the topic above.

Create modules/observability/variables.tf with input variables for: alb_arn_suffix, notification_email, log_retention_days, monthly_budget_usd, estimated_charges_threshold, and aws_region.

Task 2 — HTTP 5xx and latency alarms

Still in modules/observability/main.tf, add two aws_cloudwatch_metric_alarm resources:

1. HTTP 5xx error rate — namespace "AWS/ApplicationELB", metric_name "HTTPCode_Target_5XX_Count", dimension key "LoadBalancer" with value = var.alb_arn_suffix. Set statistic = "Sum", period = 60, evaluation_periods = 2, threshold = 5, comparison_operator = "GreaterThanOrEqualToThreshold", treat_missing_data = "notBreaching".
2. Target response time — same namespace and dimension, metric_name "TargetResponseTime", statistic = "Average", threshold = 1 (seconds). Use treat_missing_data = "notBreaching".

Both alarms must set alarm_actions = [aws_sns_topic.alerts.arn] and ok_actions = [aws_sns_topic.alerts.arn].

Task 3 — EstimatedCharges alarm with us-east-1 alias

AWS billing metrics are only published in us-east-1. You need a provider alias to reach them regardless of your default region.

1. In root versions.tf, add a second provider "aws" block with alias = "us_east_1" and region = "us-east-1".
2. In modules/observability/main.tf, add a terraform { } block declaring required_providers with aws having configuration_aliases = [aws.us_east_1]. Then add a variable "providers" is not needed — the alias is passed via the providers map when calling the module.
3. Add an aws_cloudwatch_metric_alarm for EstimatedCharges: namespace = "AWS/Billing", metric_name = "EstimatedCharges", dimensions = { Currency = "USD" }, statistic = "Maximum", period = 86400, evaluation_periods = 1, threshold = var.estimated_charges_threshold, comparison_operator = "GreaterThanOrEqualToThreshold". Set provider = aws.us_east_1 on this resource.
4. Wire the alarm's alarm_actions to the SNS topic ARN.

Task 4 — Monthly budget guard

Add an aws_budgets_budget resource in modules/observability/main.tf:

- budget_type = "COST", time_unit = "MONTHLY"
- limit_amount = var.monthly_budget_usd, limit_unit = "USD"
- One cost_notification block: threshold = 80, threshold_type = "PERCENTAGE", notification_type = "ACTUAL", comparison_operator = "GREATER_THAN", subscriber_email_addresses = [var.notification_email]

Important: aws_budgets_budget reads AWS billing data with a 24–48 hour lag. It is not suitable for real-time cost alerting. Use the EstimatedCharges alarm from Task 3 for near-real-time cost alerts; use this budget for a hard monthly ceiling.

Task 5 — Wire the module from root

In root main.tf, call the observability module and pass the us-east-1 provider alias via a providers map:

```

module "observability" {
  source = "../modules/observability"

  alb_arn_suffix      = var.alb_arn_suffix
  notification_email  = var.notification_email
  log_retention_days  = var.log_retention_days
  monthly_budget_usd  = var.monthly_budget_usd
  estimated_charges_threshold = var.estimated_charges_threshold
  aws_region          = var.aws_region

  providers = {
    aws = aws
    aws.us_east_1 = aws.us_east_1
  }
}

```

Then run:

```

terraform init
terraform apply -var-file="envs/dev/dev.tfvars"

```

After apply, check your email and click the SNS subscription confirmation link AWS sends. The subscription status in the AWS console must show "Confirmed" before you submit.

Acceptance Criteria

- modules/observability/main.tf declares aws_cloudwatch_log_group, aws_sns_topic, aws_sns_topic_subscription, aws_cloudwatch_metric_alarm × 3 (HTTP 5xx, latency, EstimatedCharges), and aws_budgets_budget
- The EstimatedCharges alarm resource sets provider = aws.us_east_1
- root versions.tf contains a provider "aws" block with alias = "us_east_1" and region = "us-east-1"
- terraform apply completes without errors
- Log group /finapi/dev (or similar) is visible in the CloudWatch console under Log groups — save a screenshot as evidence/log-group.png
- At least one alarm is visible in the CloudWatch console in ALARM or OK state — save a screenshot as evidence/alarm.png
- SNS subscription confirmation email was received and the link was clicked; subscription status in the AWS console shows Confirmed — save a screenshot as evidence/sns-confirmed.png
- Repository is named oyd-exercise-9-1 and the URL is submitted